

# Self-distillation of context — cross-checkout report

Self-distillation of context – cross-checkout comparison.

Method families compared (one row per run, best-effort artifacts):

[layerwise] 149 runs

Forward layerwise distillation (this checkout). Every block updated with uniform k-deep credit; behavioral readout teacher-sourced.

[classical-kd] 4 runs

Classical/naive KD baselines (../selfupdate\_kd): full fine-tune and LoRA KL-distillation of the top layers.

[multigpu] 17 runs

Big-model layerwise on sharded GPUs (../selfupdate\_multi): gpt-oss-120B, Qwen3.5-122B, Gemma-4, ALIA-40B.

Metric switch (owner directive, 2026-07-07): the tiny held-out general cross-entropy (log-loss) 'destruction' canary was measured to be noise-dominated and near-flat across training. It is replaced by two standard, bounded, low-variance measurements, scored in a single teacher-forced batched pass so only model LOADING costs time:

DESTRUCTION (capability retained vs the epoch-0 teacher of the base):

- ARC-Easy accuracy on a fixed cached subset (option log-likelihood)
- WikiText-2 validation perplexity (quantization-style damage)

RECALL (memorization of the original text; eval-only, never trained):

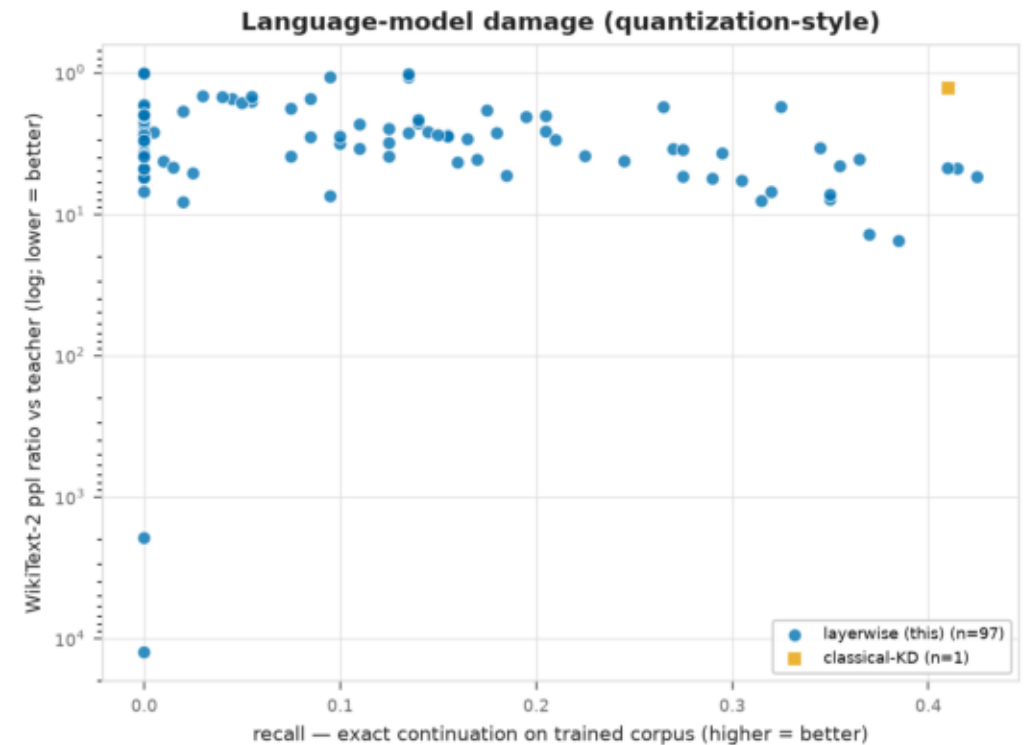
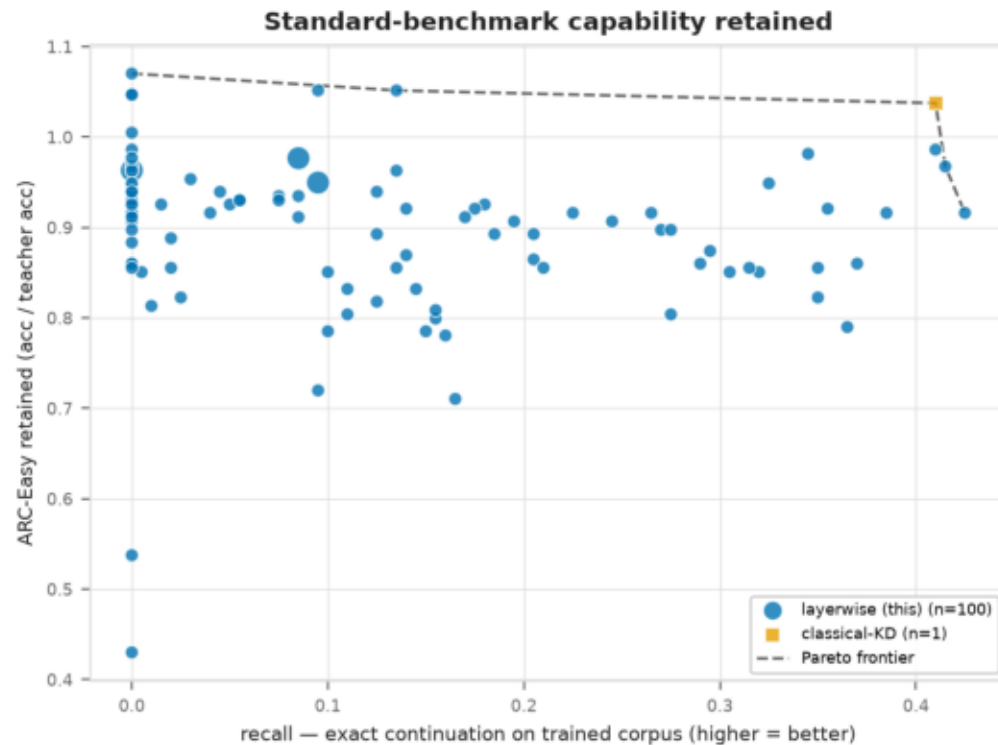
- exact continuation + interior-word cloze (Machado verse)
- next-sentence continuation + multi-word paragraph censor (Cervantes)

Coverage: 170 runs total; 101 with the new retention battery, 147 with recitation CER. Runs still needing the battery are listed on the coverage page; rerun scripts/retention\_eval.py to fill them.

Generated 2026-07-07 13:27. Backbone: runs/retention\_index.csv.

# Recall vs capability retention (bidimensional)

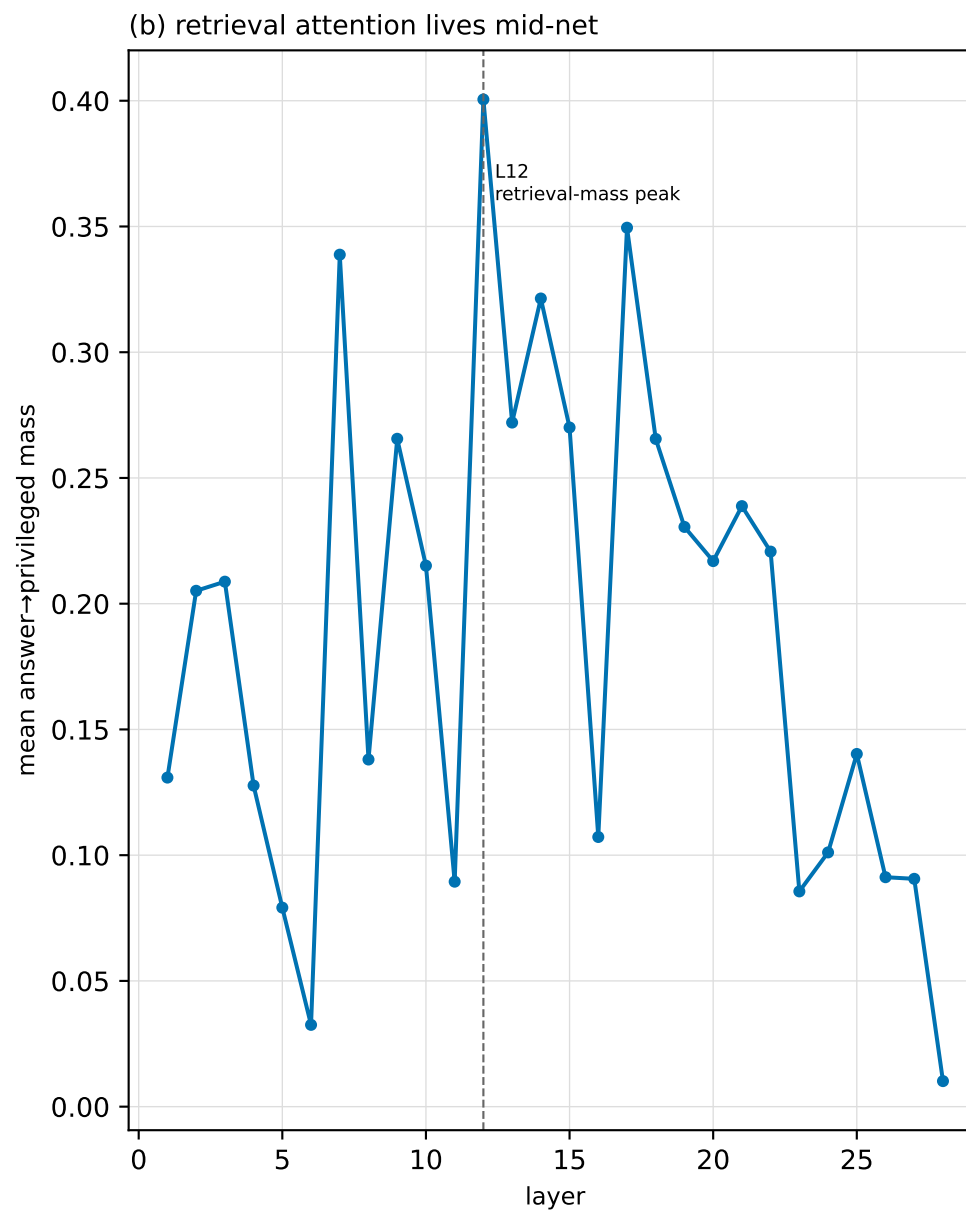
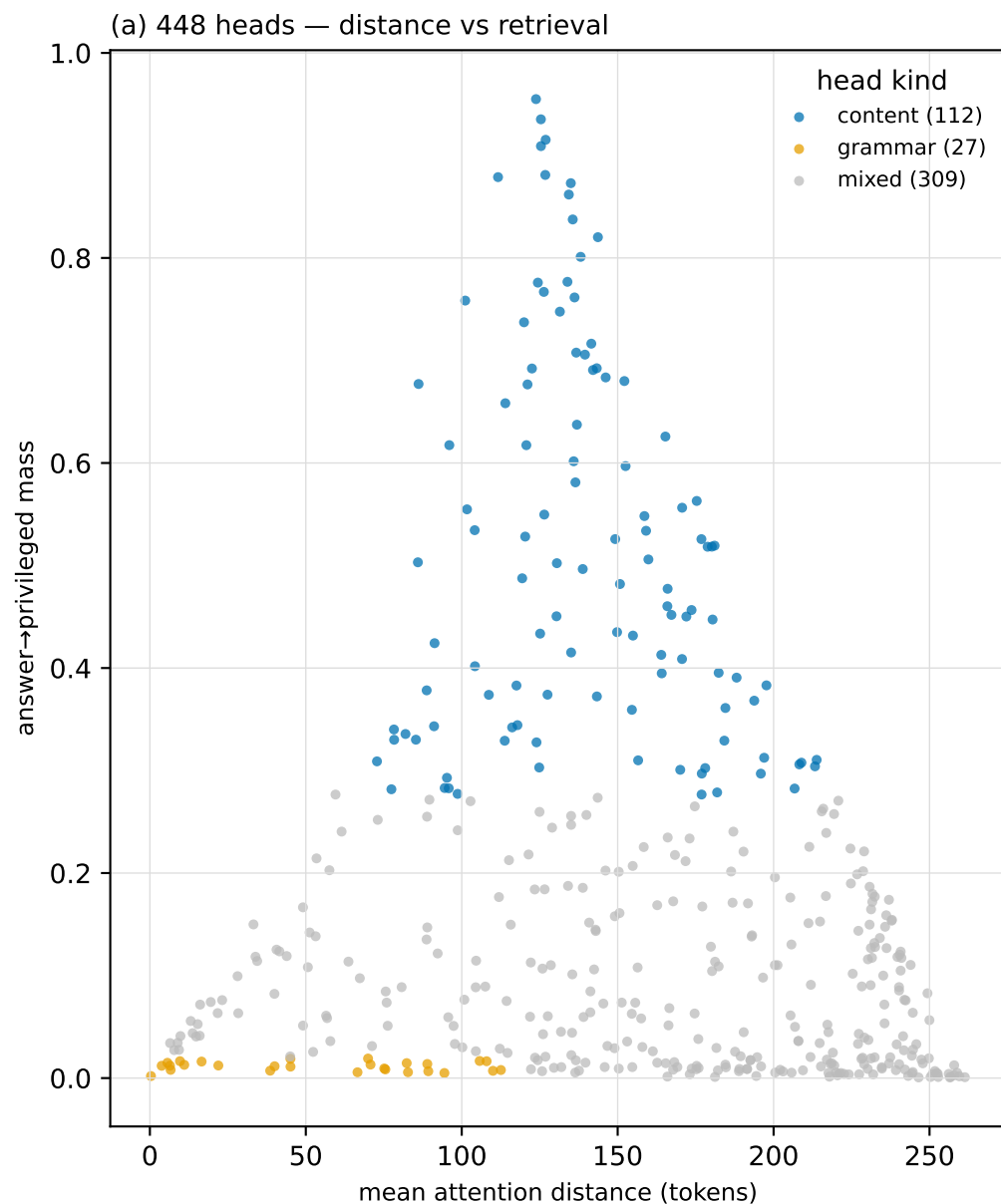
## Recall vs capability retention — endpoint Pareto across method families



101 runs with retention evidence · marker size ~ model params · colour+shape = method family (Okabe-Ito, CVD-safe)

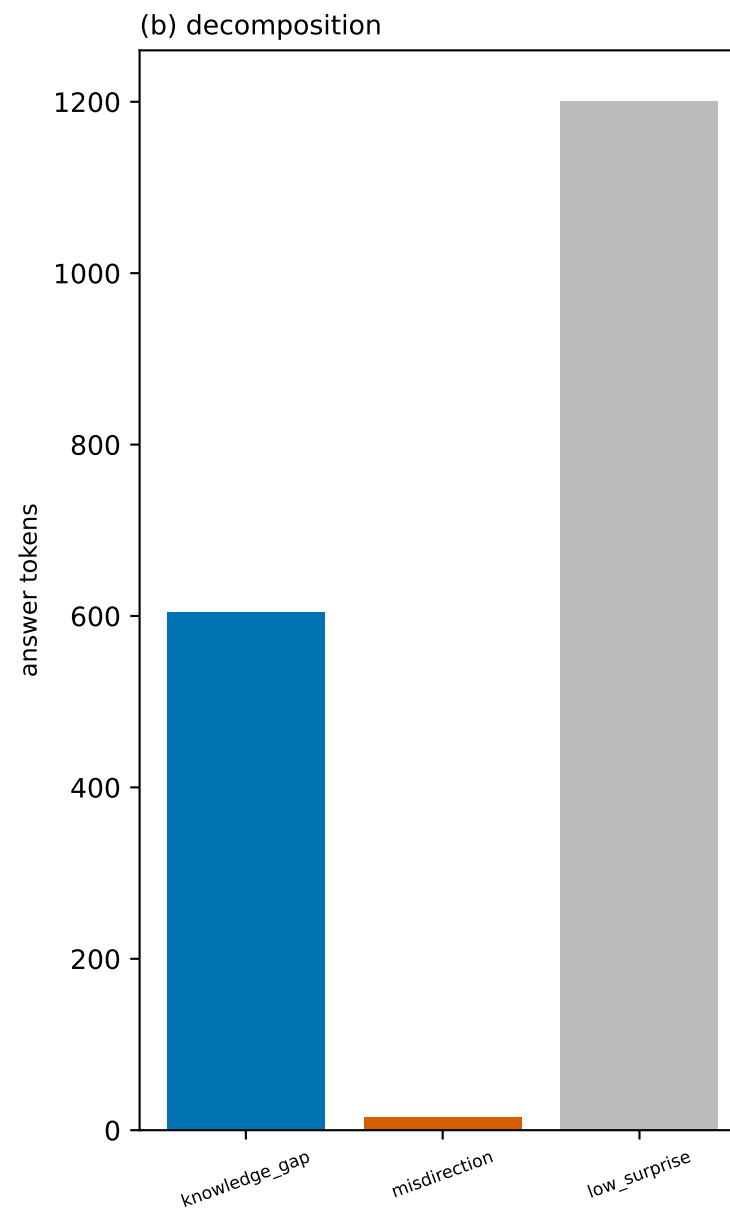
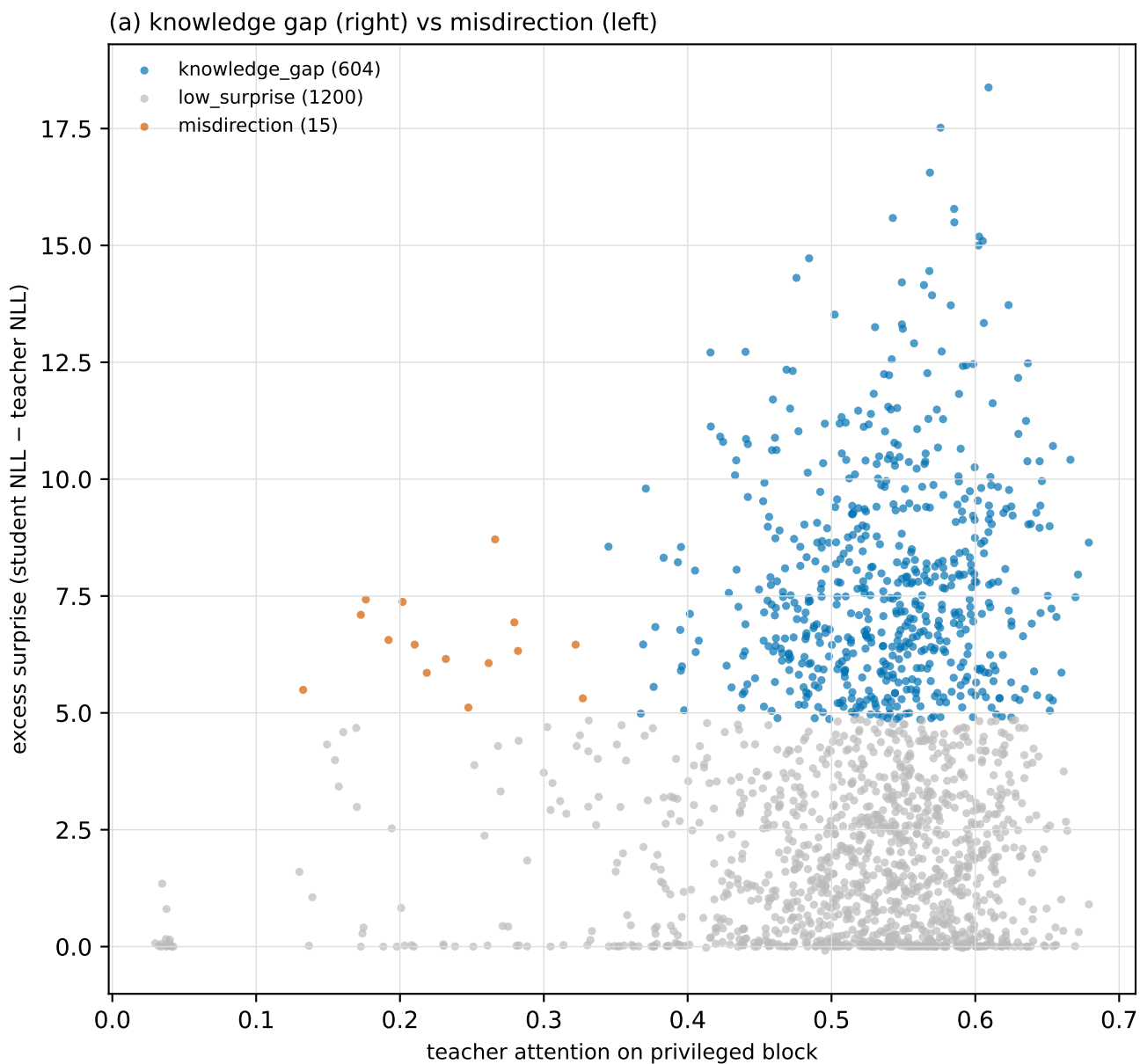
# What to memorize — attention head taxonomy (0.6B)

Teacher view (privileged passage present), answer positions. A head that reaches FAR (high distance) and lands ON the privileged block (high answer→privileged mass) is a content/insight head — it marks what the context considered worth absorbing. Local, privileged-blind heads are grammar. This is the memorization target that motivates hidden-primary (mid-net) storage, and it names what the recall probes should test.



# What to memorize — surprise decomposition (0.6B)

Surprise = student-view NLL minus teacher-view NLL on each answer token (base model, privileged block removed vs present). Footprint over content heads, structural/sink tokens masked. High surprise splits into knowledge\_gap (604; teacher attends the privileged block) and misdirection (15; teacher attends shared in-context tokens). Knowledge gaps are the memorization target; misdirection is a routing fix.



# Best retention row per method family and model

Recall: CER (lower better) and exact-match probes (higher better). Retention: ARC retained & WikiText ppl ratio vs epoch-0 teacher.

| source       | model      | run                            | recall_cer | cont_mach | cloze_mach | sensor_cerv | arc_acc | arc_retain | wiki_ratio |
|--------------|------------|--------------------------------|------------|-----------|------------|-------------|---------|------------|------------|
| layerwise    | Qwen3-0.6B | clean_q_ch1_slide8_vocab_0p6b_ | 0.980      | 0.000     | 0.009      | 0.886       | 0.573   | 1.070      | 1.98       |
| layerwise    | Qwen3-4B   | clean_machado_slide8_vocab_4b_ | 1.085      | 0.085     | —          | 0.045       | 0.725   | 0.976      | —          |
| classical-kd | Qwen3-0.6B | kd_full_0p6b_rag               | 0.863      | 0.410     | 0.991      | 0.034       | 0.555   | 1.037      | 1.29       |

# Retention coverage — remaining work

Runs WITHOUT the new retention battery (need scripts/retention\_eval.py):

```
[layerwise] 49 missing
clean_combined_slide8_vocab_14b_lora clean_combined_slide8_vocab_1p7b_rag clean_combined_slide8_vocab_8b_lora
clean_machado_slide8_vocab_14b_lora clean_machado_slide8_vocab_8b_lora clean_machado_slide8_vocab_alia40b_lora
clean_q_ch1_slide2_vocab_14b_lora clean_q_ch1_slide2_vocab_1p7b_rag clean_q_ch1_slide4_vocab_14b_lora
clean_q_ch1_slide4_vocab_1p7b_rag clean_q_ch1_slide8_lenskl_14b_lora clean_q_ch1_slide8_nmse_14b_lora
clean_q_ch1_slide8_vocab_14b_lora clean_q_ch1_slide8_vocab_1p7b_e320_rag clean_q_ch1_slide8_vocab_1p7b_rag
clean_q_ch1_slide8_vocab_8b_lora clean_q_ch1_slide8_vocab_alia40b_lora dev
lw_j_l2mse_k4_1p7b_rag lw_j_llama8b_rag lw_j_mistral7b_rag
lw_j_phi4mini_rag lw_j_qwen4b_rag lw_j_qwen8b_rag
lw_j_vocab_k2_1p7b_rag lw_j_vocab_k4_1p7b_rag lw_j_vocab_k8_1p7b_rag
lw_k_final_1p7b_rag lw_k_gptoss_rag lw_l_final_14b_rag
lw_l_final_4b_av2_rag lw_l_final_4b_av2lr_rag lw_l_final_4b_r64_rag
lw_l_final_4b_rag lw_l_final_8b_av2_rag lw_l_final_8b_rag
lw_l_tailonly_4b_rag lw_m_anchordiv_1p7b_rag lw_m_combined_1p7b_rag
lw_n_thinksel_1p7b_rag lw_p_ft4b_rag lw_r_slide8pure_1p7b_rag
lw_seq_1p7b_rag lw_tc_lora_14b_rag smoke_bucketed_mb2_gpu3
smoke_item_mb2_gpu3 xs_fisher_1p7b_rag xs_slide2_1p7b_rag
xs_slide4_1p7b_rag
```

```
[classical-kd] 3 missing
kd_lora_kl_hi_e40_v3_qwen36_27b_rag kd_lora_kl_hi_e40_v3_qwen3_30ba3b_inst2507_rag kd_lora_kl_hi_e60_v3_14b_rag
```

```
[multigpu] 17 missing
mlt_alia40b_slide2_lora mlt_gemma4_26ba4b_slide2_lora mlt_gemma4_26ba4b_slide2_ra_lora
mlt_gemma4_26ba4b_slide2_tf_lora mlt_gemma4_31b_kllocal_lora mlt_gemma4_31b_slide2_lora
mlt_gemma4_31b_slide3_lora mlt_gptoss120b_auto4_slide2_tf_lora mlt_gptoss120b_pp2_kllocal_lora
mlt_gptoss120b_pp2_slide2_lora mlt_gptoss120b_pp2_slide2_tf_lora mlt_gptoss120b_pp2_slide3_lora
mlt_gptoss20b_slide2_lora mlt_gptoss20b_slide2_ra_lora mlt_gptoss20b_slide2_tf_lora
mlt_qwen35_122b_auto4_slide2_lora mlt_qwen35_122b_gptq_auto2_slide2_lora
```